

Experiment 4.2: Design and Analysis of an FIR Low-Pass Filter Using the Hamming Window Technique

Digital Signal Processing Lab | MATLAB Simulation

1. AIM

To design and analyze a Low-Pass Finite Impulse Response (FIR) filter using the Hamming window technique in MATLAB, and observe its magnitude and phase response.

2. MATLAB CODE

```
clc;
clear;
close all;

%% Save Command Window Output
diary('Exp4_2_Output.txt');

%% Input Parameters
fp = 1000;      % Passband cutoff frequency (Hz)
fs = 4000;      % Sampling frequency (Hz)
N = 50;        % Filter order

wc = fp/(fs/2); % Normalized cutoff frequency

%% FIR Low Pass Filter using Hamming Window
b_hamm = fir1(N, wc, 'low', hamming(N+1));

disp('--- Hamming Window Coefficients ---');
disp(b_hamm);

%% Frequency Response
figure(1);
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');

%% Stop Saving Command Window
diary off;

%% Save Figure (300 DPI)
exportgraphics.figure(1), 'Exp4_2_Hamming_Window.png', 'Resolution',300);

%% Completion Message
disp(' ');
disp('=====');
disp('Experiment 4.2 Completed Successfully');
disp('Files Saved:');
disp('Exp4_2_Output.txt');
disp('Exp4_2_Hamming_Window.png');
disp('=====');
```

3. COMMAND WINDOW OUTPUT TEXT

```
--- Hamming Window Coefficients ---  
  0.0010  
    0  
-0.0013  
    0  
  0.0021  
    0  
-0.0034  
    0  
  0.0055  
    0  
-0.0084  
    0  
  0.0125  
    0  
-0.0181  
    0  
  0.0260  
    0  
-0.0379  
    0  
  0.0580  
    0  
-0.1026  
    0  
  0.3168  
  0.4995  
  0.3168  
    0  
-0.1026  
    0  
  0.0580  
    0  
-0.0379  
    0  
  0.0260  
    0  
-0.0181  
    0  
  0.0125  
    0  
-0.0084  
    0  
  0.0055  
    0  
-0.0034  
    0  
  0.0021  
    0  
-0.0013  
    0  
  0.0010
```

4. RESULT & FREQUENCY RESPONSE PLOT

